

1. What is the basic transmission rate of PDH in Indian Railways?
 - A. 1.5 Mbps
 - B. 2 Mbps
 - C. 34 Mbps
 - D. 155 Mbps
2. PDH primarily uses which multiplexing technique?
 - A. Time Division Multiplexing
 - B. Frequency Division Multiplexing
 - C. Code Division Multiplexing
 - D. Wavelength Division Multiplexing
3. Which of the following is true about PDH synchronization?
 - A. Perfectly synchronous
 - B. Almost synchronous
 - C. Asynchronous
 - D. Isosynchronous
4. PDH equipment in Indian Railways most commonly supports which interface?
 - A. E1 (2 Mbps)
 - B. E3 (34 Mbps)
 - C. STM-1 (155 Mbps)
 - D. OC-3
5. Which standard frame structure applies to 2Mb/s PDH links?
 - A. 30 voice + 2 signaling time slots
 - B. 24 voice + 1 signaling slot
 - C. 32 voice channels only
 - D. 16 user + 4 control slots
6. What is the basic SDH rate used in Indian Railway backbone networks?
 - A. 2 Mbps
 - B. 34 Mbps
 - C. 155 Mbps
 - D. 622 Mbps
7. SDH network topologies commonly implemented are:
 - A. Linear bus
 - B. Star
 - C. Ring
 - D. Mesh
8. Which of the following is a benefit of SDH over PDH?
 - A. Lower cost
 - B. Better error performance and management

- C. Mixed vendors not allowed
 - D. No support for E1 signals
9. SDH works on which synchronization concept?
- A. Isosynchronous
 - B. Synchronous**
 - C. Asynchronous
 - D. Almost synchronous
10. The standard payload mapping in SDH for carrying 2 Mbps signals is called:
- A. VC-12**
 - B. VC-4
 - C. VC-3
 - D. VC-1
11. What is the typical core diameter of multimode optical fibre?
- A. 125 microns
 - B. 50-62.5 microns**
 - C. 9 microns
 - D. 100 microns
12. For backbone transmission in Indian Railways, which fibre type is preferred?
- A. Multimode
 - B. Single-mode**
 - C. Plastic fibre
 - D. Free-space optics
13. Which wavelength has the lowest attenuation in optical fibre?
- A. 850 nm
 - B. 1310 nm
 - C. 1550 nm**
 - D. 1625 nm
14. Router is a _____ device
- A. Layer 1
 - B. Layer 2
 - C. Layer 3**
 - D. Layer 4
15. In Indian Railway telecom, fibre loss per km is generally expected as:
- A. <0.5 dB/km for single-mode at 1550 nm**
 - B. 1 dB/km for multimode
 - C. 1.5 dB/km for plastic fibre
 - D. >2 dB/km for old fibre
16. Optical fibres used in Indian Railways should comply with which standard?

A. IEEE 802.3

B. ITU-T G.652D

C. ISO 9001

D. GSM-R

17. The maximum distance between optical repeater stations for SDH is generally:

A. 5 km

B. 20 km

C. 40 km

D. 80 km

18. Which is NOT a fibre-optic connector type?

A. SC

B. LC

C. RJ45

D. ST

19. OTDR is used in optical fibre networks mainly for:

A. Transmitting data

B. Measuring optical power

C. Locating faults by reflectometry

D. Amplifying signals

20. "Dark fibre" in Indian Railway networks means:

A. Fibre in use

B. Spare/leased fibre, not transmitting light

C. Old fibre

D. Fibre with attenuation

21. While cable laying bending radius should not be sharper than _____ D where D is the diameter of cable.

A. 10 B. 20 **C. 30** D. 40

22. Indian Railways primarily uses which protocol for network addressing?

A. UDP

B. TCP/IP

C. X.25

D. HDLC

23. Which organization manages the IP address allocation for Indian Railways?

A. TRAI

B. RailTel Corporation

C. DOT

D. IIT-Kanpur

24. Line side impedance matching of VF transformer:
A 1120 ohm **B 470 ohm**
25. Diameter of cladding of single mode fibre
A. 125 micro meter
B. 10 micro meter
C. 8 micro meter
D. 100 micro meter
26. Which cable is used for interfacing with EI system.
A. Multimode fibre cable.
B. Singlemode fibre cable
27. RailTel backbone typically connects which types of railway sites?
A. Colonies only
B. Stations, junctions, and control offices
C. Private sidings
D. Signal posts only
28. What does "ASN" refer to in Railways' Internet domain context?
A. Autonomous System Number
B. Addressing Sector Number
C. Absolute Subnet Number
D. Access String Node
29. The primary function of IP addressing in railway telecommunication is:
A. Labelling equipment
B. Facilitating communication in the network
C. Creating employee records
D. Security audits
30. Which Indian Railways entity is associated with the AS38799 network?
A. RailTel Corporation of India Ltd.
B. IRCTC
C. Centre For Railway Information Systems
D. Konkan Railway